

# Project Master File

*AI Engineer Assessment: Document Understanding, Grounded Drafting, and Edit Improvement*

## 1. Executive Summary

This project is a local-first document understanding and grounded drafting system. It ingests messy legal-style documents, scans, screenshots, handwritten notes, PDFs, and partially legible records; extracts usable text and structured data; retrieves source-grounded evidence; generates first-pass drafts; and improves future outputs based on operator edits.

| Decision Area    | Chosen Approach                                                                | Reason                                                                   |
|------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Architecture     | Local-first pipeline with BYOK paid fallback                                   | Demonstrates practical cost control, privacy awareness, and reliability. |
| Local OCR        | PaddleOCR/Tesseract adapter with confidence scoring                            | Fast and auditable baseline with word/line confidence.                   |
| Local LLM/Vision | Ollama-compatible local model adapter                                          | Keeps MVP runnable without mandatory cloud dependency.                   |
| Paid fallback    | Gemini Vision + Claude Vision BYOK adapters                                    | Used only when local extraction confidence or validation fails.          |
| Grounding        | Chunked document store + citations + evidence spans                            | Prevents unsupported generation and lets reviewers inspect evidence.     |
| Quality metrics  | OCR confidence, coverage, retrieval support, citation completeness, edit delta | Makes quality visible and actionable.                                    |

## 2. Assessment Requirements Coverage

| Rubric Area                    | Implementation Response                                                                                                            | Evidence in Submission                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Document Processing            | Accept PDFs, images, scans, screenshots, handwritten notes; preprocess; OCR; extract text and structured fields; score confidence. | Spec sections 3, 5, 6; test cases DP-01 to DP-08. |
| Retrieval and Grounding        | Chunk documents, embed locally, retrieve relevant source spans, attach citations to draft output, block unsupported claims.        | Spec sections 7, 8; test cases RG-01 to RG-07.    |
| Draft Quality                  | Generate chosen draft types with clear structure, disclaimers, citations, and unknowns.                                            | Spec section 9; acceptance tests DG-01 to DG-05.  |
| Improvement from Edits         | Capture operator edits, classify edit types, store reusable lessons, update prompts/templates.                                     | Spec section 10; tests IE-01 to IE-05.            |
| Code Quality and System Design | Modular adapters, provider interface, deterministic tests, error handling, configuration separation.                               | Spec sections 12-16; playbook.                    |
| Documentation and Clarity      | Master file, agent spec, build journal, run/playbook, architecture notes, sample test plan.                                        | This package.                                     |

## 3. Product Goal

- Turn messy source documents into grounded, reviewable drafts without hallucinated facts.
- Run locally by default to reduce cost and protect sensitive materials.
- Escalate only failed or low-confidence pages/fields to BYOK paid models.

- Show users an accuracy/quality dashboard before they rely on the draft.
- Learn from accepted operator edits without storing raw sensitive data unnecessarily.

## 4. User Workflow

| Step | User/Operator Action   | System Behavior                                                                                   |
|------|------------------------|---------------------------------------------------------------------------------------------------|
| 1    | Upload document bundle | Stores files, creates document job, fingerprints pages.                                           |
| 2    | Select draft type      | Options: review summary, case fact summary, notice-related summary, checklist, first-pass memo.   |
| 3    | Local processing       | Runs preprocessing, OCR, structure extraction, confidence scoring.                                |
| 4    | Quality decision       | If local quality passes, continue; if not, ask for BYOK paid fallback or use configured API keys. |
| 5    | Grounded generation    | Retrieves evidence and creates draft with citations and unsupported/unknown section.              |
| 6    | Operator edits         | User edits draft; system records edit patterns and future prompt improvements.                    |

## 5. Fallback Policy

- Default provider order: local OCR/local model -> Gemini BYOK vision -> Claude BYOK vision.
- Fallback is per page or per field, not all-or-nothing, to control cost.
- Fallback triggers: OCR confidence below threshold, missing required fields, retrieval has no evidence, image-only page with poor extraction, or validation inconsistency.
- When no API key exists, the system surfaces the degraded quality state and lets the operator continue with warnings.

## 6. Quality Metrics Dashboard

| Metric                  | Formula / Meaning                                        | Green | Amber  | Red  |
|-------------------------|----------------------------------------------------------|-------|--------|------|
| OCR Confidence          | Weighted average confidence over recognized tokens/lines | >=90  | 75-89  | <75  |
| Page Coverage           | Pages with usable extraction / total pages               | >=95% | 80-94% | <80% |
| Field Completeness      | Required extracted fields present                        | >=95% | 80-94% | <80% |
| Retrieval Support       | Draft claims with cited evidence                         | >=90% | 75-89% | <75% |
| Unsupported Claim Count | Claims without source support                            | 0     | 1-2    | >2   |
| Edit Delta              | Percent of draft changed by operator                     | <=20% | 21-40% | >40% |

## 7. Suggested MVP Scope

- One web app with upload, job status, extraction view, draft view, edit capture, and quality dashboard.
- Local adapters for OCR, embeddings, vector retrieval, and local LLM.
- BYOK settings page for Gemini API key, Claude API key, and provider/model order.
- Five draft templates: review summary, case fact summary, notice summary, document checklist, first-pass memo.

- Golden dataset of synthetic legal-style samples for deterministic testing.

## 8. Open Questions Before Coding

- Which draft type should be the default demo output for the office submission?
- Should the UI be desktop-only for MVP, or also mobile-friendly?
- Which local runtime is acceptable on the target machine: Docker, Python virtualenv, Node, or all three?
- Are documents expected mostly in English, Bengali, or mixed language?
- Should paid fallback happen automatically once BYOK keys exist, or require user confirmation per job?

## Appendix A: Recommended Repository Structure

- /app
- /api
- /components
- /lib/providers
- /lib/ocr
- /lib/retrieval
- /lib/quality
- /lib/drafting
- /lib/edits
- /tests
- /fixtures
- /docs
- .env.example
- README.md
- SPEC.md